MIL-I-60513 (MU)
13 June 1966
SUPERSEDING
FA-PD-MI-2357, Rev 1
17 Jan 1961

MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, DELAY, M45 ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, Delay, M45, Assembly, loaded with Cartridge, Initiator, Delay, M114, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION
Military
- MIL-A-2550 - Ammunition and Special Weapons, General Specification For

STANDARDS
Military
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions
- MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
- MIL-STD-1167 - Ammunition Data Card
- MS-28741 - Hose Assembly, Detachable End Fitting, Medium Pressure

DRAWINGS
Munitions Command
- F8596101 - Initiator, Cartridge Actuated, Delay, M45, Assembly
- C8796826 - Container, Ammunition, Fiber M375 for Initiators, Delay, M4, M6A1, M12 or M33 (see 5.1)
- C8796827 - Box, Packing, Ammunition for Initiators, Delay, M4, M6A1, M12 or M33 in Fiber Container M375 (see 5.1)
- IEL-8596101 - Index of Inspection Equipment List for Initiator, Delay, M45, Assembly

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:40Z
Check the source to verify that this is the current version before use.

2.1 (cont)

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Firing pin release.- The firing pin of the assembled initiator, less safety pin, shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin 3/4 inch, maximum.

3.3 Initiator pin security.- Application of an axial load of 200 to 220 pounds to the initiator pin of the assembled initiator shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin.

3.4 Breakaway torque.- The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.5 Residual magnetism.- The initiator shall not deflect the indicator of the compass more than three degrees in either direction when passed in front of the compass at a distance of six inches.

3.6 Functional.

3.6.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in table I. Pressure values shall be recorded at the end of three feet of aircraft hose, size no. 4, MS-28741.

Table I

Temperature	Pressures (psi)	Ignition Delay (see 6.2)
-65° to -70° F	1300 to 2200	4.20 seconds, max
65° to 75° F	1300 to 2200	3.0 ± 0.3 seconds
160° to 165° F	1300 to 2200	2.30 seconds, min

3.6.2 The initiator shall not fail to fire.

3.6.3 The initiator shall function without mechanical failure (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:40Z
Check the source to verify that this is the current version before use.

3.7 Workmanship.- Requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.7.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-414, as applicable.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 Examination.- One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in table II. All nonconforming material shall be rejected. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of table II. Four power (4X) magnification may be used for inspection of such defects as burr and foreign matter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:40Z
Check the source to verify that this is the current version before use.

4.2.2 (cont)

Table II

Lot Size	Sample Size	Major Accept	Major Reject	Minor Accept	Minor Reject
1 - 300	100% examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Initiator, Delay, M114, Dwg D8596110, covering a detail of Dwg F8596101.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Foreign matter	Visual
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Dwg F8596101, less Dwgs D8596110, D8593313, D8593114 and B8359536.

CRITICAL	
1. Mixed initiators	Visual
2. Initiator pin not in cocked position	Visual
3. Safety pin missing or improperly engaged	Visual
MAJOR	
101. O-ring missing or not lubricated	Visual
102. Thread damaged	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.3 Assembly, Dwg F8596101, less Dwgs D8593314 and B8359536.

CRITICAL	
1. Cartridge loose	Visual/Manual
2. Cartridge retainer loose	Visual/Manual
3. Metering groove of cartridge retainer missing	Visual
4. Elongated hole of cartridge retainer missing	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:40Z
Check the source to verify that this is the current version before use.

4.2.2.3 (cont)

Defect	Method of Inspection
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.4 Assembly, Dwg F8596101, less Dwg B8359536.

CRITICAL	
1. Cartridge retainer location	Gage
2. Foreign matter in gas port	Visual
MAJOR	
101. Length from cap boss face to gas port face, max	Gage
102. Protrusion of firing pin, max	Gage
103. Set screw missing, loose or above flush	Manual/Visual
MINOR	
None defined.	

4.2.2.5 Assembly, Dwg F8596101.

CRITICAL	
1. Base of firing pin housing not in complete contact with face of cartridge	Manual
2. Any part missing, except as otherwise classified	Visual
3. Improper assembly	Visual
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Height of cap, max	Gage
102. Initiator pin not free to rotate	Manual
103. Safety pin loose or deformed	Manual/Visual
104. Set screw not covered with barrier material	Visual
MINOR	
201. Total length, max	Measure
202. Location of labels incorrect	Visual
203. Shipping cap missing	Visual
204. Protective finish inadequate or defective	Visual

4.2.2.6 Fiber Container, Dwg C8796826.

CRITICAL	
None defined.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:40Z
Check the source to verify that this is the current version before use.

4.2.2.6 (cont)

	Defect	Method of Inspection
MAJOR		
	None defined.	
MINOR		
201.	Gap between body and cover, max	Manual
202.	Contents loose	Manual
203.	Metal ends, loose or distorted	Manual
204.	Cut or damaged through all impregnated layers	Visual
205.	Marking missing, incorrect or illegible	Visual
206.	Tape incomplete or badly wrinkled	Visual
207.	Length of tear tab, min	Visual
208.	Color of tape incorrect	Visual

4.2.2.7 Packing, Box, Dwg C8796827.

CRITICAL		
	None defined.	
MAJOR		
	None defined.	
MINOR		
201.	Contents loose	Manual
202.	Hardware missing, loose, broken or improperl engaged	Visual/Manual
203.	Strapping missing, loose, broken or improperly assembled	Visual/Manual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking misleading or unidentifiable	Visual
207.	Number of units incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.3.

4.2.3.1 Firing pin release.- The initiator pin of each subassembly shall be subjected to the firing-pin release test. Failure of any subassembly to comply with the requirement of 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Initiator pin security.- The initiator pin of each subassembly shall be subjected to the initiator-pin security test. Failure of any unit to comply with the requirement of 3.3 shall be cause for rejection of the subassembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:40Z
Check the source to verify that this is the current version before use.

4.2.3.3 Breakaway torque.- Each chamber and cap assembly shall be subjected to the breakaway torque application. Failure of any assembly to comply with the requirement of 3.4 shall be cause for rejection of the assembly.

4.2.3.4 Residual magnetism.- Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirement of 3.5 shall be cause for rejection of the assembly.

4.2.3.5 Functional tests.- The following tests shall be performed by the Government:

4.2.3.5.1 Ballistics.- Twenty-one units shall be tested as shown in figure 2, to determine compliance with the requirements of 3.6.1, 3.6.2 and 3.6.3. The units shall be temperature conditioned as provided in table III and 4.3.5.1.1. The variables sampling plan of MIL-STD-414, example C2 or C3 as applicable, for the AQL listed in table III shall be used for each sample of seven units. Failure of any unit to comply with the requirements or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

Table III

Temperature	Sample Size	AQL
-65° to -70° F	7	0.65%
65° to 75° F	7	0.65%
160° to 165° F	7	0.65%

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8596101.

4.3 Test methods and procedures.

4.3.1 Firing pin release.- A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the applicable fixture which registers the amount of pull.

4.3.2 Initiator pin security.- A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the applicable fixture which registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.3.3 Breakaway torque.- A torque wrench set at the torque requirement specified in 3.4 shall be used in conjunction with a fixture to determine the adequacy of the joint. The chamber shall be secured in an air vise or holding fixture. The torque shall be applied to the cup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:40Z
Check the source to verify that this is the current version before use.

4.3.4 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator six inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading as shown in figure 1. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90-degree rotation of the initiator.

4.3.5 Functional.

4.3.5.1 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within five minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned at the original temperature.

4.3.5.1.1 Temperature conditioning.- A standard M45 Initiator, loaded with a dummy M114 Cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with dwgs C8796826 and C8796827.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

Custodian:

Army - MU

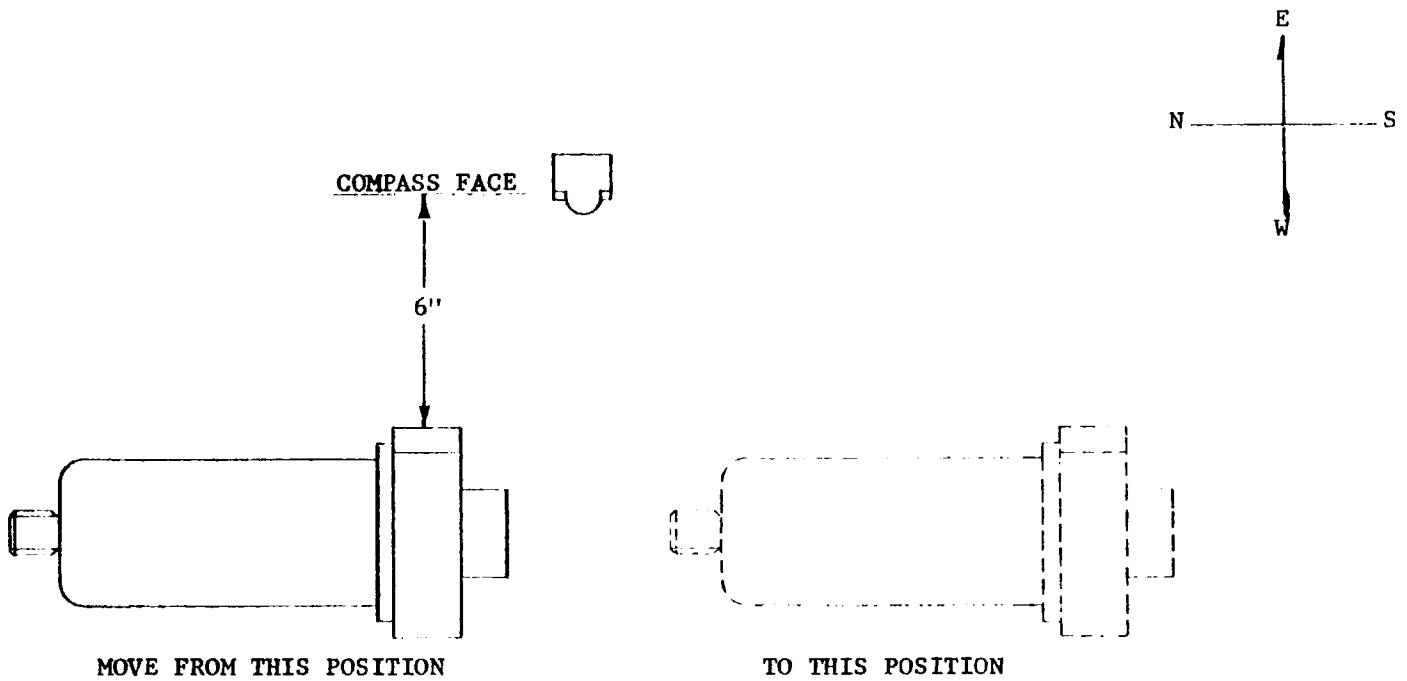
Preparing activity:

Army - MU

Project No. 1377-A039

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:40Z
Check the source to verify that this is the current version before use.

PILOT'S STANDBY COMPASS, TYPE AN-5766



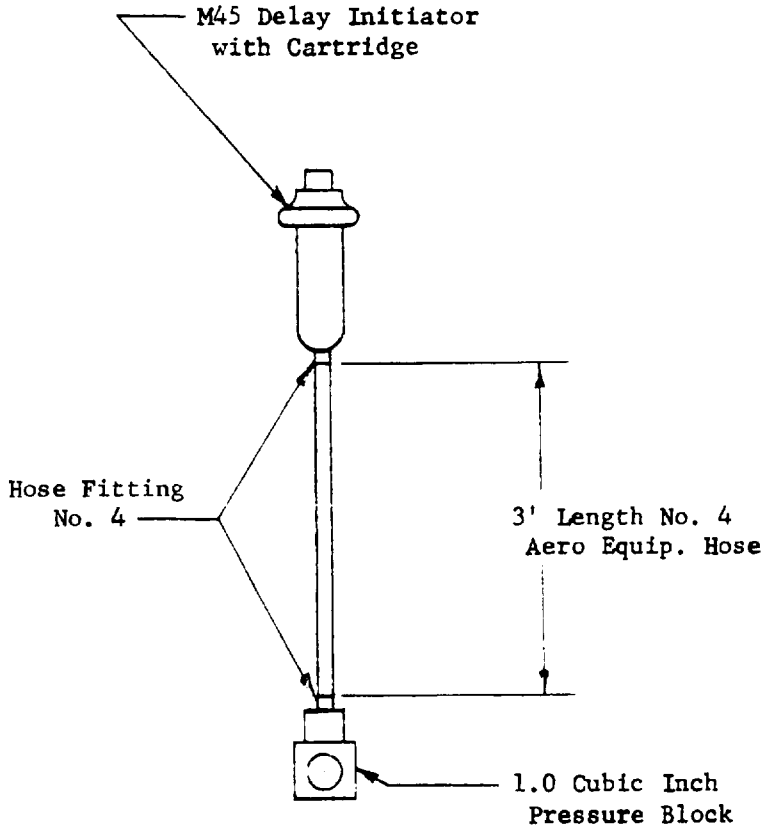
PLAN VIEW

NOTE:
INITIATOR AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN INITIATOR, CARTRIDGE ACTUATED, DELAY, M45

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:40Z
Check the source to verify that this is the current version before use.



PERFORMANCE TEST LAYOUT SKETCH

TOP VIEW

FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:40Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

Frankford Arsenal
Philadelphia, Pa. 19137

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-N1100
Philadelphia, Pa. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:40Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:40Z
Check the source to verify that this is the current version before use.